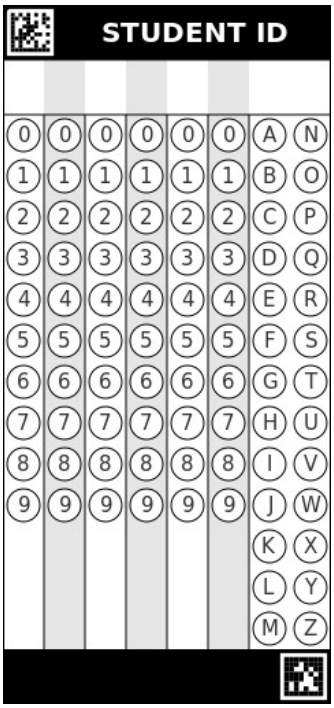
HWA CHONG INSTITUTION
Preliminary Examination
Higher 2

CANDIDATE NAME								
CT GROUP	2	5	S			**TUTOR NAME**		
CENTRE NUMBER	S					**INDEX NUMBER**		



PHYSICS

9478 / 02

Paper 2 Structured Questions

15 September 2026

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

- Write your **name**, **CT group, centre number** and **index number** on all the work you hand in.
- **Write** and **shade** your Student ID using a 2B pencil.
  Your Student ID is xxxxxxx, where your HCI email is xxxxxxx@student.hci.edu.sg
- Use a dark blue or black pen. You may use a pencil for any diagrams or graphs.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** remove any staple; Do **not** detach any pages of this question booklet.
- You may use an approved calculator.

INFORMATION

Answer all questions.

The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use		
1		**/ 8**
2		**/ 6**
3		**/ 4**
4		**/ 4**
5		**/ 7**
6		**/ 9**
7		**/ 7**
8		**/ 7**
9		**/ 23**
Deductions		
P2		**/ 75**

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_o = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_o = 8.85 \times 10^{-12}$ F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}$ m F$^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/ by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T / K = T / °C + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

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displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^{\,2} - x^2\right)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 Figure 1.1 shows the variation with the speed v of the force F_D driving a car forward and the resistive force F_R that opposes the motion of the car.

The mass of the car is 1200 kg.

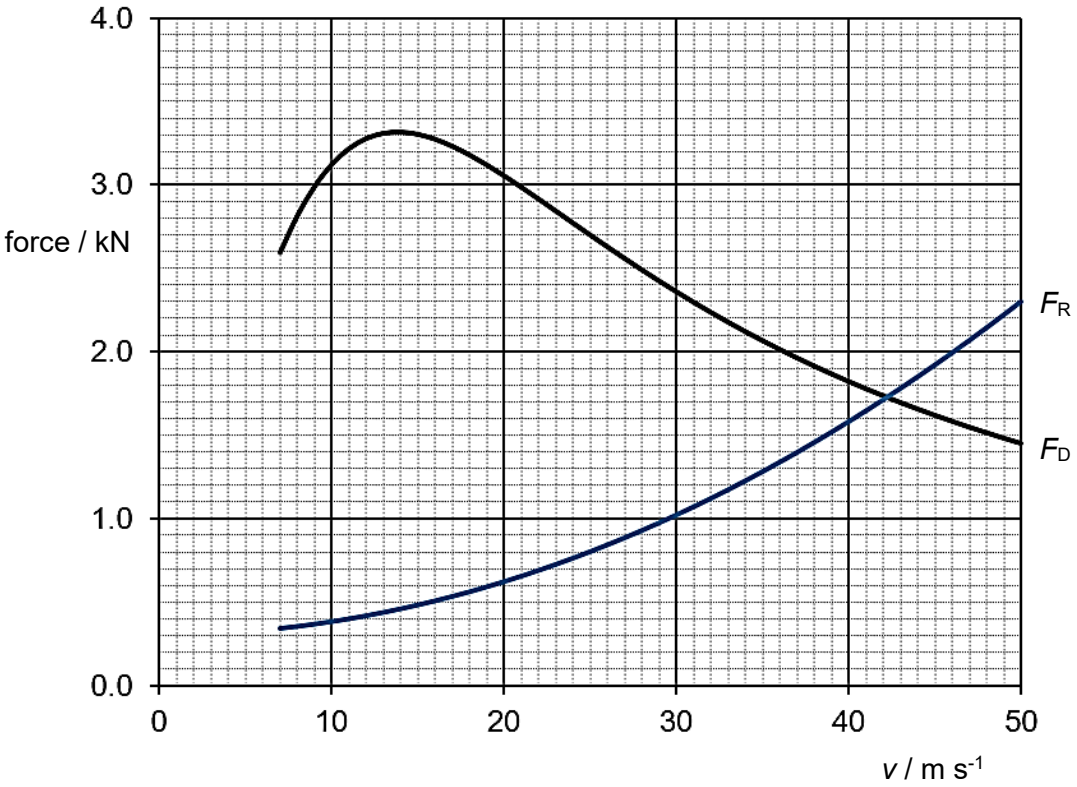


Figure 1.1

(a) Use Figure 1.1 to determine, for the car travelling along a horizontal road at **maximum resultant force,**

(i) the speed at which the car is travelling,

speed = ... m s^{-1} [1]

(ii) its maximum acceleration,

acceleration = ... m s^{-2} [2]

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(iii) the rate of increase of kinetic energy of the car.

rate of increase of kinetic energy = ... W [2]

(b) The car now travels up a slope inclined at an angle to the horizontal.

(i) Explain why, the maximum speed the car can have is less than that when it is travelling on level road.

.....

.....

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..... [2]

(ii) On Figure 1.1, sketch the variation with speed *v* of the total force opposing the motion of the car up the slope.

Label this graph as **S**.

[1]

[Total: 8]

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2 A ball is projected from point A, as shown on Figure 2.1.

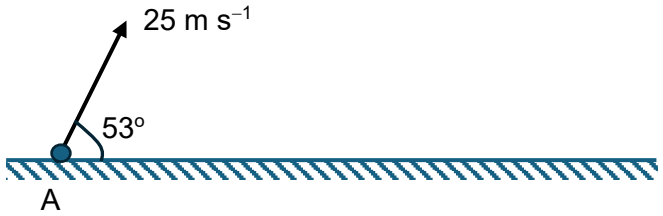


Figure 2.1

The initial velocity of the ball is 25 m s^{-1} at an angle of 53° to the horizontal. The mass of the ball is 0.20 kg.

(a) At a point P on its way to its highest point, the ball's velocity is at an angle of 45° to the horizontal.

(i) Show that the magnitude of the vertical component of the velocity of the ball at point P is 15 m s^{-1}.

[1]

(ii) Calculate the time it takes for the ball to travel from point A to point P.

time = ……..…… s [1]

(b) Calculate the kinetic energy of the ball at a height of 8.0 m.

kinetic energy = …… J [2]

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(c) On Figure 2.2, sketch the variation with height of the kinetic energy of the ball.
No numerical values are needed.

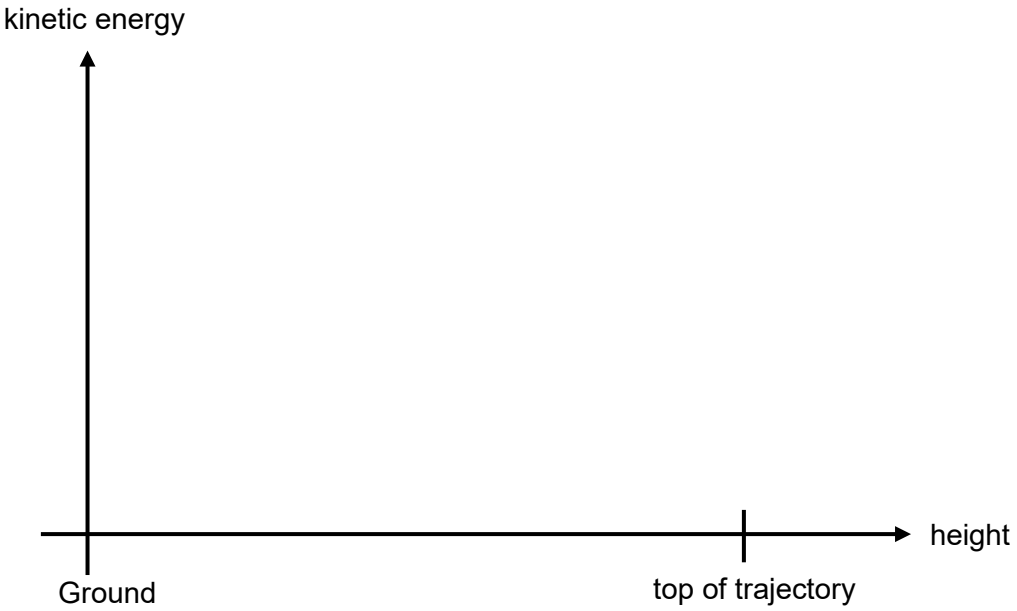


Figure 2.2

[2]

[Total: 6]

3 A beam of unpolarised light of intensity I_o is incident normally on a set-up consisting of three polarisers P_1, P_2 and P_3, arranged as shown in Figure 3.1.

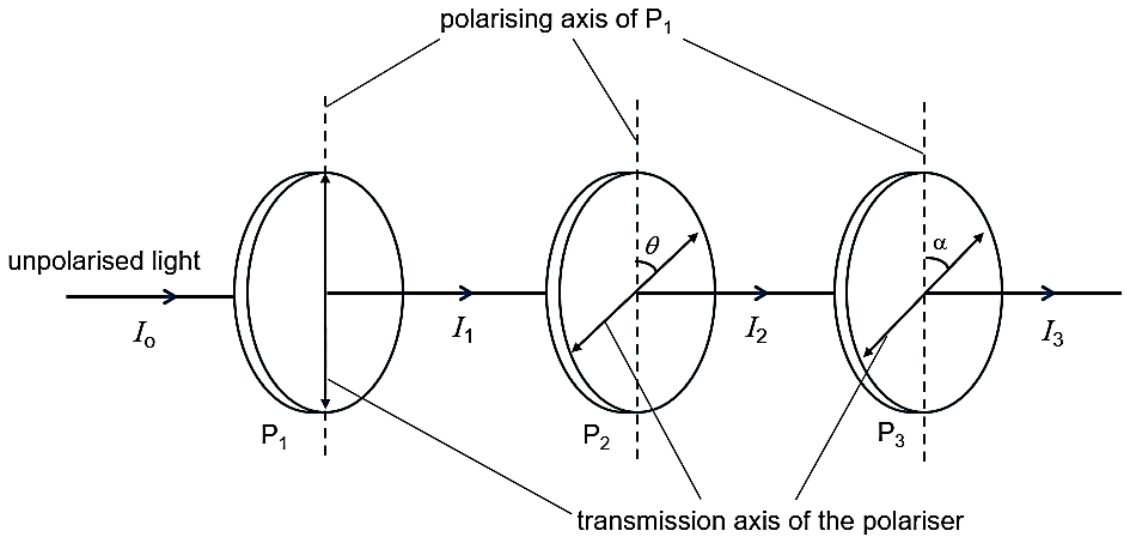


Figure 3.1

The light is plane polarised vertically after passing through P_1. The transmission axis of P_2 is at an angle θ to the vertical and the transmission axis of P_3 is at an angle α to the vertical.

The intensity of the light transmitted through P_1, P_2 and P_3 are I_1, I_2 and I_3 respectively.

(a) Calculate the transmitted intensity I_3 in terms of I_o when $\theta = 25°$ and $\alpha = 90°$.

I_3 = ……..…….. [2]

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(b) Angle α of polariser P_3 is kept fixed at 90° while angle θ of polariser P_2 is varied.

In Figure 3.2, sketch the variation of the amplitude of transmitted light after passing through P_3 with θ. Numerical values on the vertical axis are not needed.

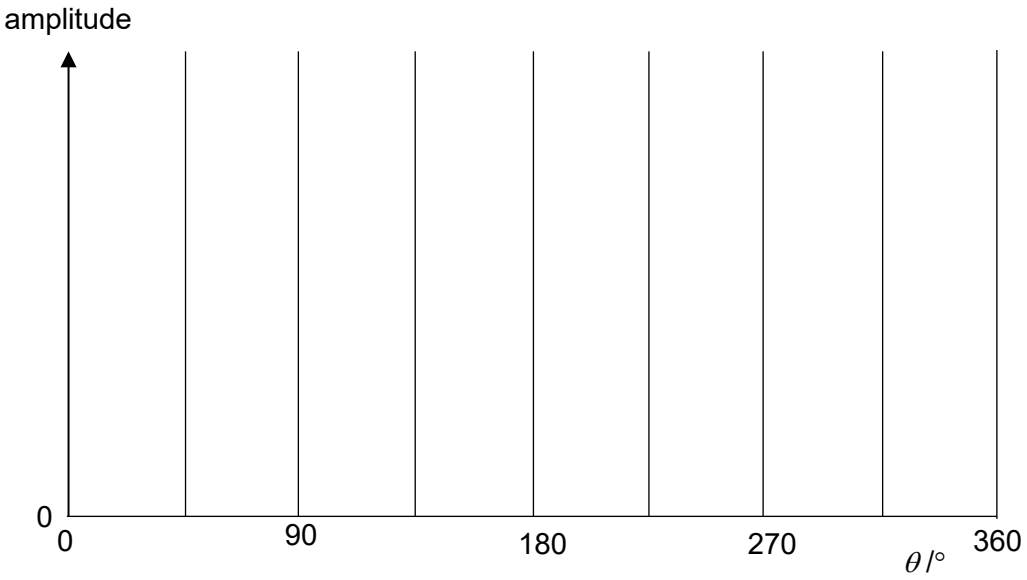


Figure 3.2

[2]

[Total: 4]

4 A receiver aerial is placed between a point source emitting microwaves at a single fixed wavelength of 140 mm and a reflective metal sheet, as shown in Figure 4.1.

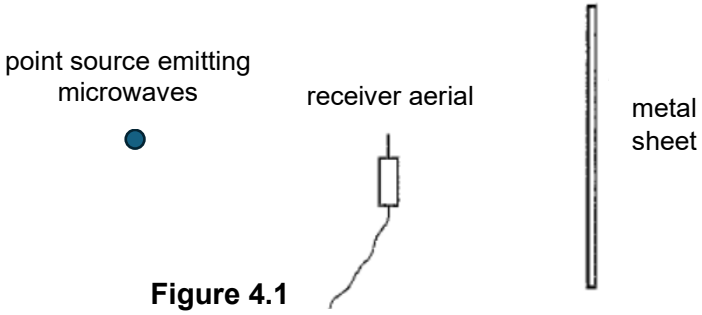


Figure 4.1

The receiver aerial is fixed in position. The metal sheet is moved away from the receiver aerial toward the right. The detected voltage at the receiver aerial varies with the distance moved by the sheet as shown in Figure 4.2.

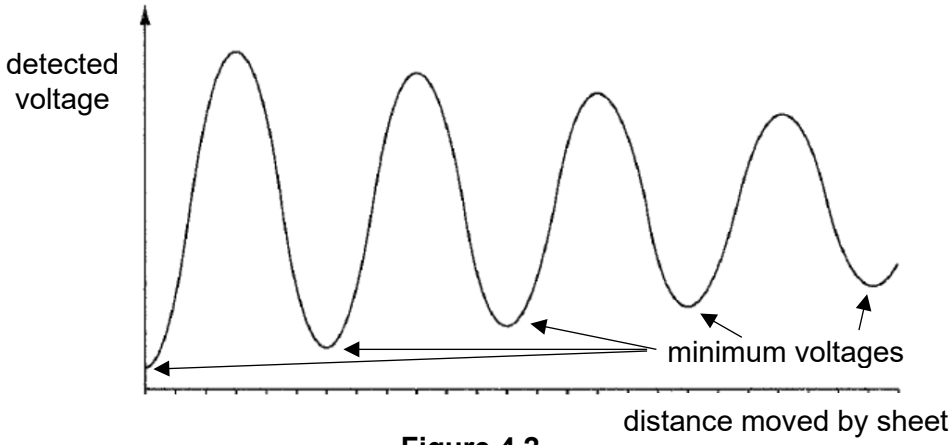


Figure 4.2

(a) Explain why the minimum voltage detected at the receiver aerial gets larger with distance moved by the sheet.

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..... [3]

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(b) State the distance moved by the metal plate between consecutive detection by the aerial receiver of minimum voltages.

distance = ... mm [1]

[Total: 4]

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5 A small conducting sphere of mass of 0.12 g is suspended from a very long insulating thread between metal plates M and N, separated by 2.0 cm as shown in Figure 5.1.

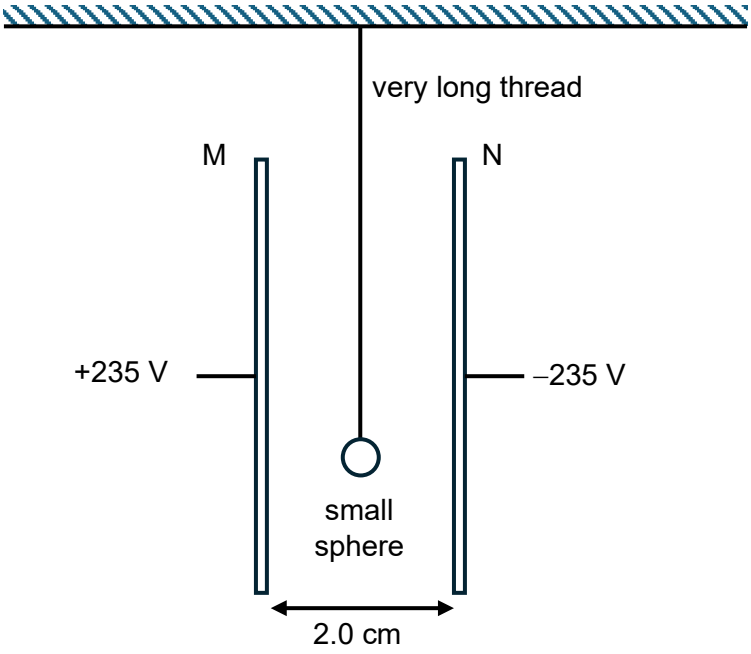


Figure 5.1 (not to scale)

Plates M and N are connected to a high-voltage power supply and are held at potentials of 235 V and -235 V, respectively.

After being given an initial displacement, the sphere moves to plate M, touches it and acquires a charge of 58 pC.

It then moves towards plate N, touches it and the horizontal oscillation process repeats itself.

Assume that the initial velocity of the sphere is negligible when moving away from the plates.

(a) Calculate the electrostatic force on the sphere as it moves towards plate N.

force = …….. N [2]

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(b) Calculate the acceleration of the sphere.

acceleration = ... m s^{-2} [1]

(c) Determine the frequency of oscillation of the sphere. Assume the size of the sphere is negligible compared to the separation of the plates.

frequency = ... Hz [3]

(d) A student claims that the oscillation is simple harmonic.

Explain whether the student is correct.

.....

.....

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..... [1]

[Total: 7]

6 Figure 6.1 shows a circuit that produces rectification of an alternating input voltage.

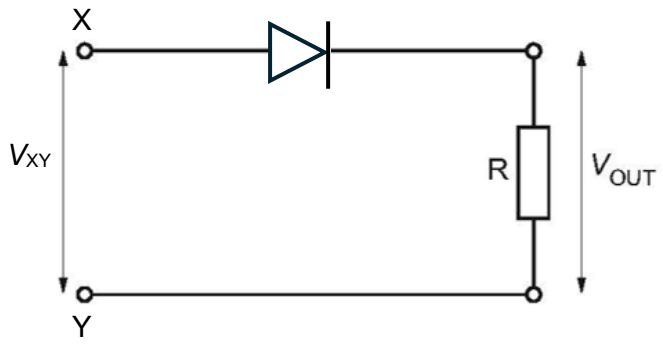


Figure 6.1

(a) The input voltage V_{XY} is the potential of X with respect to Y and is sinusoidal. V_{OUT} is the rectified voltage across resistor R.

The variation of V_{XY} with time t has amplitude V_0 and period T, as shown in Figure 6.2.

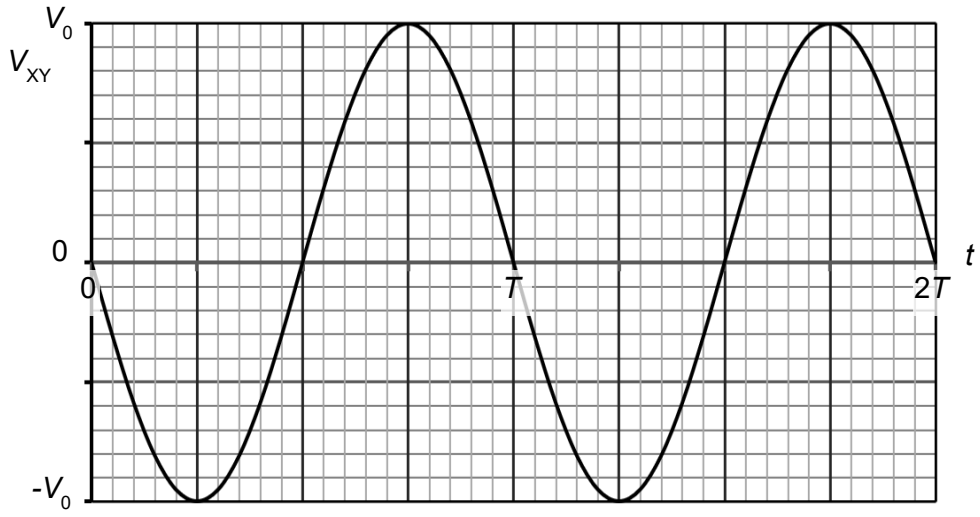


Figure 6.2

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(i) On Figure 6.3, sketch the variation with time t of the power P in the resistor between $t = 0$ and $t = 2T$.

Label the peak power P_0 in the resistor on your sketched graph.

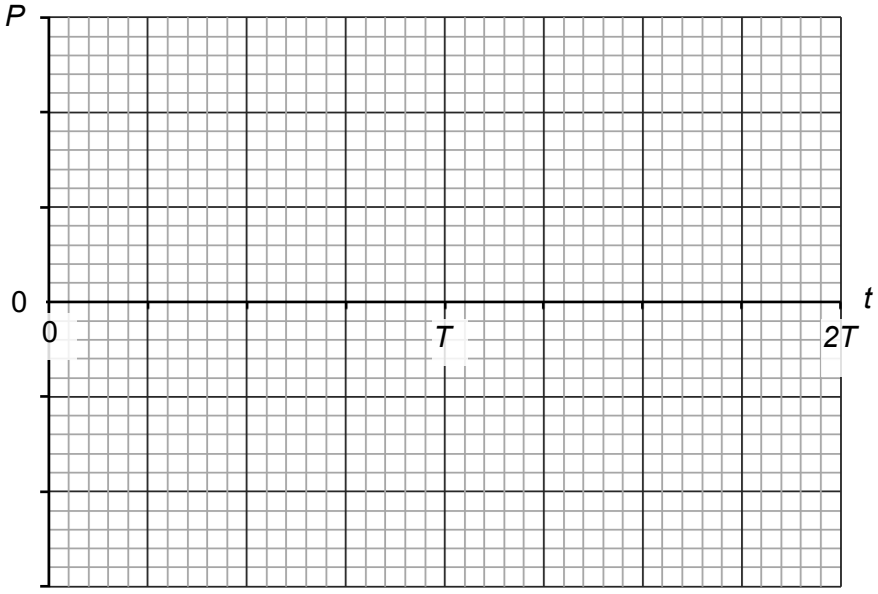


Figure 6.3

[2]

(ii) Use the answer in **(a)(i)** to explain why the mean power in the resistor is $\frac{1}{4}P_0$.

.....

.....

.....

..... [2]

(iii) The root-mean-square (r.m.s.) value of V_{XY} is 12 V.

Use the information in **(a)(ii)** to show that the r.m.s. value of V_{OUT} is 8.5 V.

[2]

(b) To smooth the output voltage, a capacitor is connected in parallel to the resistor in Figure 6.1.

During each discharge cycle, the minimum value of the smoothed output voltage is 0.60 V_0,

(i) Figure 6.4 shows a duplicate of the graph of V_{XY} with time t.

On Figure 6.4, sketch the variation with time t of the smoothed rectified output voltage V_R across the resistor R between $t = 0$ and $t = 2T$.

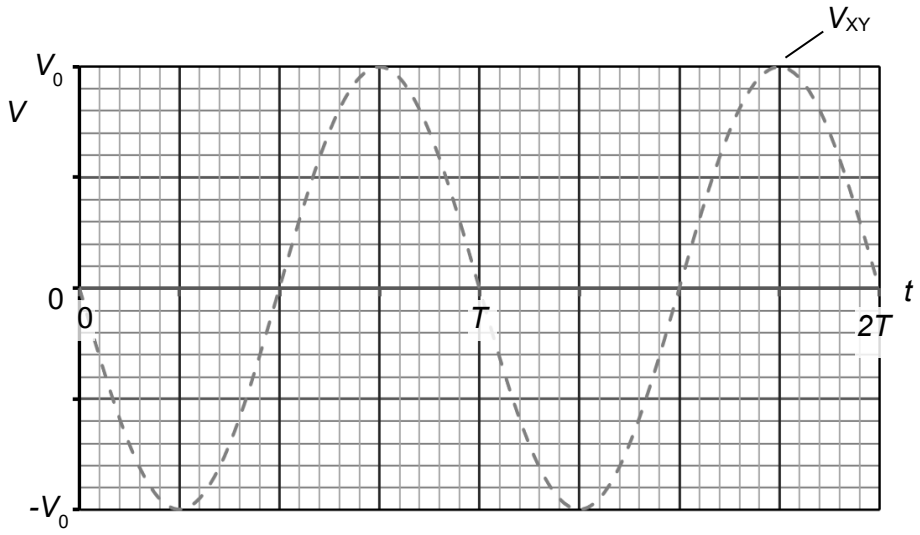


Figure 6.4

[1]

(ii) Determine the time constant of the smoothing circuit in terms of the period T.

time constant = ... [2]

[Total: 9]

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7 An electron is trapped in a one-dimensional infinite potential well of width L along the x-direction. It can be modelled as a standing wave described by the following wavefunction:

$$\psi(x) = \begin{cases} A\sin\left(\dfrac{n\pi x}{L}\right) & 0 \le x \le L \\ 0 & \text{everywhere else} \end{cases}$$

where n is associated with different possible energy states (n = 1, 2, 3, …)

(a) For the n = 2 state,

(i) Sketch in Figure 7.1 the wave function $\psi(x)$ of the electron.

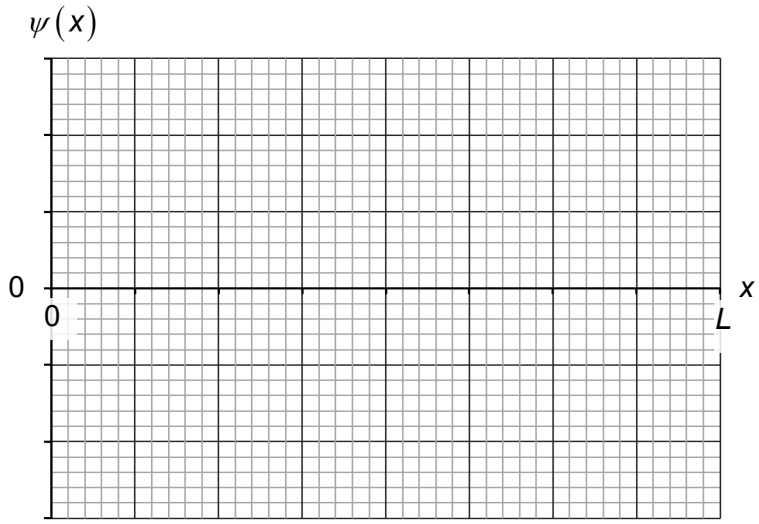


Figure 7.1

[1]

(ii) Sketch in Figure 7.2 the probability density function $|\psi(x)|^2$.

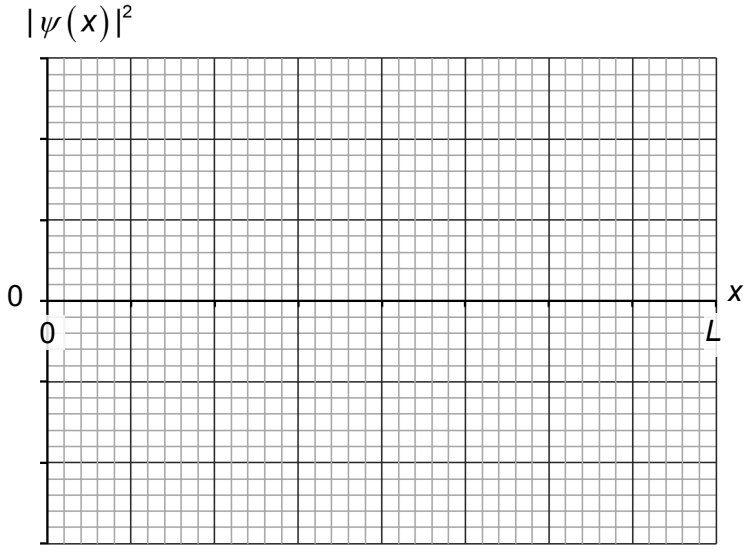


Figure 7.2

[1]

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(b) **(i)** Explain what is meant by a *normalised wave function*.

.....

..... [1]

(ii) Express the constant A in terms of L so that the wave function is normalised.

A = ... [1]

(c) A semiconductor quantum dot can be modelled as an infinite potential well.

A particular semiconductor quantum dot has an effective length L of 2.2 nm. An electron trapped in it has an effective mass of 0.15 times the mass of a free electron.

Determine the wavelength of the photon emitted when an electron in the first excited state drops to the ground state.

wavelength = ... m [3]

[Total: 7]

8 **(a)** State the experimental evidence that suggests that the process of radioactive decay is spontaneous.

……

……

…… [1]

(b) The radioactive isotope iodine-131 ($^{131}_{53}\mathrm{I}$) undergoes beta-decay. It has a decay constant of 8.6×10^{-2} day^{-1}.

In nuclear medicine, iodine-131 is injected into or swallowed by a patient to target and destroy cancerous cells in the thyroid gland.

(i) Suggest why alpha radiation is not suitable for destroying targeted cancerous thyroid cells.

……

……

…… [1]

(ii) A sample of blood contains 1.2×10^{-9} g of iodine-131.

Determine, for this sample of blood,

1. the activity of the iodine-131,

activity = …… Bq [3]

2. the time for the activity of the iodine-131 to be reduced to 1/50 of the activity calculated in **(ii)1**.

time = …… s [2]

[Total: 7]

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9 Read the passage below and answer the questions that follow.

The Sun is the star at the heart of our solar system. Its gravity holds the solar system together, keeping everything – from the biggest planets to the smallest particles of debris – in its orbit. The connection and interactions between the Sun and Earth drive the seasons, ocean currents, weather, climate, radiation belts and auroras.

The Sun has produced a near-constant radiant power of 4.0×10^{26} W for around 5 billion years and is only halfway through its supply of hydrogen fuel. Hydrogen is fused into helium in the Sun's core in a three-stage process known as the proton-proton cycle.

Stage 1 Deuterium Formation**:** Two hydrogen nuclei collide to form a deuterium nucleus releasing a positron and a neutrino

Stage 2 Helium-3 Formation**:** The deuterium nucleus quickly captures another proton, creating a helium-3 nucleus and releasing a gamma-ray photon

Stage 3 Helium-4 Formation: Two helium-3 nuclei collide, resulting in one helium-4 nucleus and releasing protons back into the core

Fusion requires extremely high temperatures to give the hydrogen nuclei (protons) sufficient energy to overcome electrostatic repulsion. But even at the high temperature of 1.5×10^7 °C in the core (Figure 9.1), the average energy of the protons is not enough to allow fusion. This provided a puzzle for physicists which was finally explained by a process called quantum tunnelling.

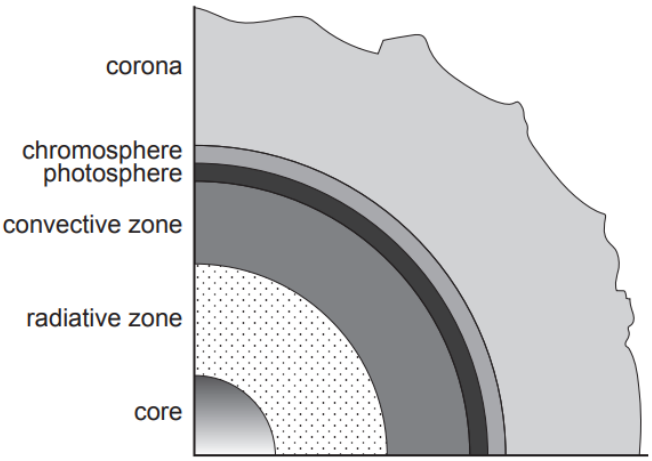


Figure 9.1 Cross section of the Sun (not to scale)

The solar wind is a continuous stream of charged particles (plasma), mainly protons and electrons, escaping the Sun's hot outer atmosphere (corona) at speeds of 500 km s^{-1}.

The Earth's magnetosphere deflects most of the solar wind (Figure 9.2). For particles that sneak through, they spiral in toward the poles, where they can contribute to the glowing lights of the auroras (Figure 9.3) when they smash into particles in the atmosphere.

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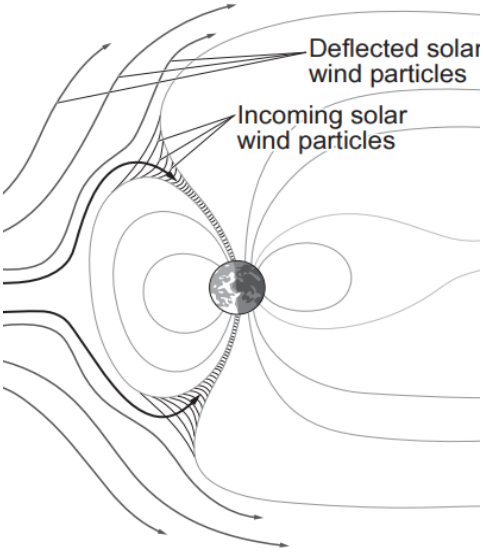


Figure 9.2 Earth's magnetic field deflecting much of the solar wind



Figure 9.3 The aurora borealis seen in Norway

Solar flares on the other hand are associated with coronal mass ejections, rapid release of plasma and ions of a typical total energy of 10^{20} to 10^{25} J. The ejected material travels through space at relativistic speeds and carries its own magnetic field which can temporarily compress Earth's magnetic field. The resulting change in Earth's magnetic field strength can burn out transformer stations in the power grid leading to blackouts. Solar research is hence an important field.

One of the most ambitious missions is the Parker Solar Probe, launched in 2018 to perform a series of passes over the Sun in a gradually changing orbit. The closest approach that the Parker Probe has made to the Sun is 1.4 x 10^{10} m. The power from the Sun at small distances is very large and is expected to raise the temperature of the probe to extremely high values. We can get an idea of the temperature it may reach if we make the simplifying assumption that the probe is in thermal equilibrium – that it radiates as much power per square metre as it absorbs. If this is the case, the probe would be behaving as a "black body radiator" and we can use Stefan's Law to estimate its temperature.

Stefan's Law states that the power absorbed or emitted per unit area by a black body in thermal equilibrium = σT^4, where T is the thermodynamic temperature and σ = 5.7 x 10^{-8} W m^{-2} K^{-4}.

(a) The equation for stage 1 of the proton-proton cycle is shown below:

$$ {}_{1}^{1}\mathrm{H} + {}_{1}^{1}\mathrm{H} \rightarrow {}_{1}^{2}\mathrm{H} + {}_{+1}^{0}\mathrm{e} + {}_{0}^{0}\nu $$

Write the equations for the subsequent stages:

Stage 2: ……

Stage 3: ……..……..……..…… [2]

(b) (i) Assuming the protons behave like an ideal gas, show that the average kinetic energy of a proton in the core of the Sun is 3.1×10^{-16} J.

[1]

(ii) Two protons need to come within 10^{-14} m of each other to fuse.

Show that on average, the protons do not have enough energy to fuse.

Explain your reasoning clearly.

[3]

(c) (i) Explain what is meant by the wave function in quantum mechanics.

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(ii) Quantum tunnelling is a quantum mechanical phenomenon in which an object such as an electron or atom passes through a potential energy barrier that, according to classical mechanics, should not be passable due to the object not having sufficient energy to pass or surmount the barrier.

Figure 9.4(a) shows a particle of energy E passing through a potential energy barrier U_0.

Figure 9.4(b) gives the wave function of the particle in the corresponding regions of space.

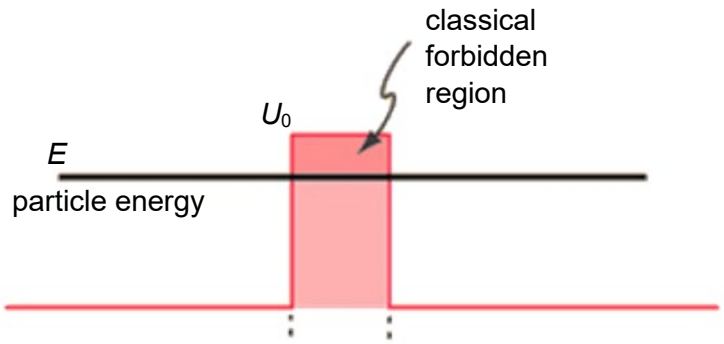


Figure 9.4(a)

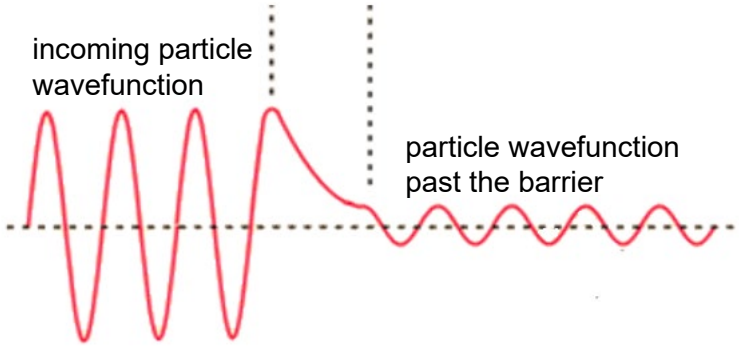


Figure 9.4(b)

With reference to Figure 9.4(b),

1. explain how the amplitude of the wave function shows the particle can exist beyond the potential barrier.

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..... [2]

2. by considering the wavelength of the wave function beyond the potential barrier, comment on the energy of the particle after quantum tunnelling.

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(d) Consider an electron in the solar wind travelling at an angle of 25° to the direction of the Earth's magnetic field of 40 μT as shown in Figure 9.5.

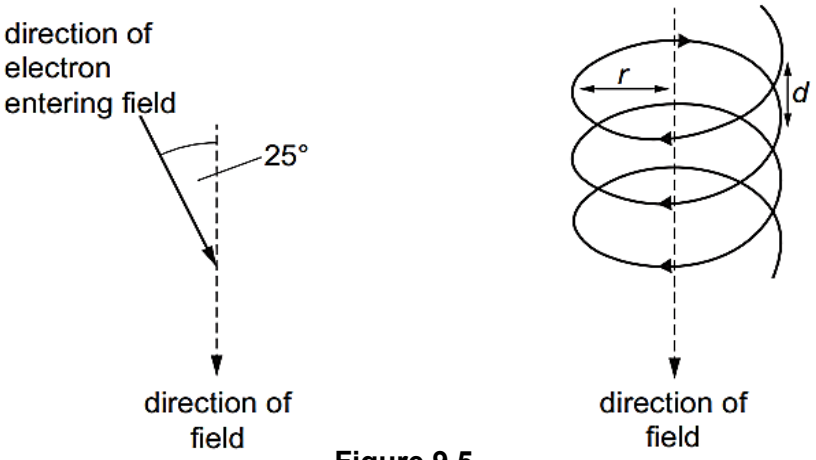


Figure 9.5

(i) By considering the horizontal and vertical components of its velocity, calculate the vertical distance *d* the electron travels between a full rotation around the field line.

distance *d* = ……m [4]

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(ii) State and explain one change to the path if it were a proton instead of an electron coming in at the same angle.

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(iii) Explain the mechanisms that cause the atoms in the atmosphere to produce light (aurora) when smashed by the charged particles.

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..... [2]

(e) **(i)** Estimate the temperature of the Parker Solar Probe at its closest distance to the Sun. Assume the Sun to be a point source of power.

temperature = ... K [3]

(ii) Suggest why missions investigating the Sun such as the Parker Solar Probe can have an importance beyond scientific discovery.

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